

system voltages, but the days of continued scaling are gone. Compounding the issue, each process node shrink means additional static power consumption for transistor-heavy SRAM-based FPGAs. This is due to worsening problems, such as quantum tunneling and sub-threshold leakage, which create real challenges for devices targeted at power-conscious applications. With leakage worsening, static power is becoming the designer's biggest concern (see Fig. 1).

As opposed to the SRAM cell structure that incurs substantial leakage and requires power-consuming configuration memory, flash-based cells have no leakage path and thus 1000 times lower leakage per cell than SRAM. A flash-based solution can consume up to 16 times lesser power per input/output (I/O). Assuming a design requires 100 I/Os, a 120-I/O device can consume as little as 5 μ W versus the roughly 60 μ W of competing solutions.

To address some of these power concerns, several suppliers of SRAM-based FPGAs claim to offer single-chip, flash-based solutions. These hybrid solutions are merely combinations of flash memory components with the underlying SRAM FPGA technology—either integrated with the FPGA die into a single package or, alternatively, stacked or placed side-by-side. Unfortunately, the FPGA array is still volatile and subject to the power drawbacks associated with these types of devices. With these solutions, the embedded flash memory blocks control only the initial configuration of the devices during power-up.

Single-chip, flash-based FPGAs help solve the low-power problem. These do not require an external configuration device (boot PROM or microcontroller) to support device programming at every power-up cycle.

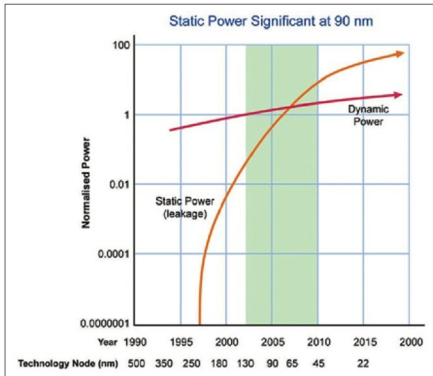


Fig. 1: With leakage worsening, static power consumption of FPGAs is becoming the designer's biggest concern. Each process node shrink means additional static power consumption for transistor-heavy SRAM-based FPGAs

Certainly, both the silicon-in-package and the multichip package hybrid approaches overcome some of the limitations of traditional SRAM-based solutions by providing a smaller footprint, a minor reduction in power consumption, and small advances in power-up time and security. But these are only incremental improvements over their pure SRAM-based peers.

The Flash-based FPGA solution

Single-chip, flash-based FPGAs help solve the low-power problem. These do not require an external configuration device (boot PROM or microcontroller) to support device programming at every power-up cycle. The 'live at power-up' feature eliminates the need for an external device to assist in system power-up. Removing the additional parts required by SRAM-based FPGAs not only reduces the board space and system power consumption, but also increases reliability, simplifies inventory management and lowers the

total system cost by as much as 70 per cent compared to similar SRAM-based FPGA solutions.

As complete flash-based solutions, these devices are augmented by software that enables power-conscious design, including power-driven layout and advanced power-analysis capabilities, permitting users to minimise the power consumption of their systems. Since each watt that is conserved reduces the system's operating costs, the deployment of cost-effective power management solutions at the enterprise level saves huge amounts of money as well as energy and generates significant environmental benefits.

High-density flash FPGAs can deliver up to 1700 times lesser static power than high-density SRAM FPGAs. When compared to low-density FPGAs and CPLDs, flash FPGAs can deliver up to 25 times less power.

Design for reality

To know how this works in practice, let's examine the medical device sector,